

* Linear Programming

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► Motivation: What is Operations Research (OR)?

- Modern organizations deal with competing divisions, scarce resources, and complex trade offs
- Definition: Operations Research (OR) is the applications of scientific and analytical methods to decision-making and resource management.

The General Mathematical Model of OR

maximize or minimize: **Objective function** → Performance measure (profit, cost)
subject to: **Constraints** → Physical, financial or logical limits

* What is Linear Programming

↳ simplest and most widely used analytical model within Operations Research

Linear Programming (LP) is the problem of optimizing (maximizing or minimizing) a linear function subject to linear constraints. A variety of practical problems are the following

- Nutrition
- Transportation
- Product Planning
- Finance

→ military term referred to plans or schedule, for training, logistical supply, or deployment of men.

and many more areas can be modeled as linear programs.

In a general linear programming problem, we are given a cost vector $C = [C_1 \ C_2 \ \dots \ C_n]^T$, we seek to minimize or maximize a linear objective function

$$C^T X = \sum_{i=1}^n C_i x_i$$

over all n -dimensional vectors $x = [x_1 \ x_2 \ \dots \ x_n]^T$ subject to a set of linear equality and inequality constraints.

$$(LP) \quad \begin{array}{ll} \min & C^T x \\ \text{s.t.} & Ax \geq b \\ & x \geq 0 \end{array}$$

where

- $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^{n \times 1}$: vector of decision variables
- $c = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^{n \times 1}$: vector of objective coefficients (cost or profit per unit)
- $b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \in \mathbb{R}^{m \times 1}$: resource capacities or requirements
- $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}$: Constraint matrix, each a_{ij} represents the consumption (or contribution) of item i per unit of activity j .

Alternatively we can write the formulation as

$$\text{minimize } c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

$$\text{subject to: } a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n \geq b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n \geq b_2$$

$$\vdots$$

$$a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n \geq b_m$$

$$x_1, x_2, \dots, x_n \geq 0$$

$$\text{minimize } [c_1 \ c_2 \ \dots \ c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\text{subject to: } \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \geq \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$x_1, x_2, \dots, x_n \geq 0$$

Prototype example: The Diet Problem

We have a limited budget, we would like to find the most economical mix of food items subject to providing sufficient daily nutrition. The available food items are peanut butter, bananas, and chocolate, and the cost per serving is 20 cents, 10 cents, and 15 cents, respectively. The amount of nutrients of fat, carbohydrates, and protein per serving of peanut butter is 130 grams, 51.6 grams, and 64.7 grams. For bananas, the amounts of these nutrients per serving are 1 gram, 51 grams, and 2 grams, whereas for chocolate the amounts per serving are 12 grams, 22 grams, and 2 grams. Suppose it is decided by a dietician that you need at least 35 grams of fat, at least 130 grams of carbohydrates, and at least 76 grams of protein daily.

Goal:

minimize: cost of food items used

subject to: food items must provide enough

- fats
- carbohydrates
- protein

Formulation: Let x_1 = servings of peanut butter
 x_2 = servings of bananas
 x_3 = servings of chocolate

Our cost vector is $c = [0.20 \ 0.10 \ 0.15]^T$. The diet problem linear program is

$$\begin{array}{ll} \min & 0.20x_1 + 0.10x_2 + 0.15x_3 \quad (\text{cost}) \\ \text{s.t.} & 130x_1 + 1x_2 + 12x_3 \geq 35 \\ & 51.6x_1 + 51x_2 + 22x_3 \geq 130 \\ & 64.7x_1 + 2x_2 + 2x_3 \geq 76 \\ & x_1, x_2, x_3 \geq 0 \end{array} \quad \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \text{constraints}$$

A vector x satisfying all of the constraints is called a **feasible solution** or **feasible vector**. The set of all feasible solutions is called the **feasible set** or **feasible region**.

* Standard form Problems

A linear programming problem of the form

$$\begin{aligned} & \text{minimize } c^T x \\ & \text{subject to } Ax = b \\ & x \geq 0 \end{aligned}$$

is said to be in standard form.

* Reduction to Standard form

a) **Elimination of free variables:** Given an unrestricted variable x_j , we replace it by $x_j^+ - x_j^-$, where $x_j^+, x_j^- \geq 0$

b) **Elimination of Inequality constraints:** Given

$$\sum_{j=1}^n a_{ij} x_j \leq b_i$$

introduce a **slack variable** $s_i \geq 0$ so that we have

$$\sum_{j=1}^n a_{ij} x_j + s_i = b_i$$

Similarly for $\sum_{j=1}^n a_{ij} x_j \geq b_i$, introduce a **surplus variable** $s_i \geq 0$, so that we have

$$\sum_{j=1}^n a_{ij} x_j - s_i = b_i$$

Remark: $\max \{ c^T x \mid Ax \leq b, x \geq 0 \} = -\min \{ -c^T x \mid Ax \leq b, x \geq 0 \}$.

Negating a set of numbers flips its ordering. For any function $f(x)$ over a feasible set S :

$$\sup_{x \in S} f(x) = -\inf_{x \in S} (-f(x))$$

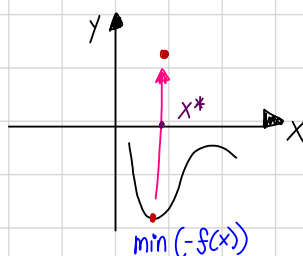
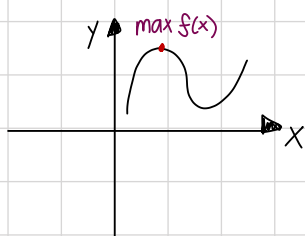
If the problem has an achievable optima, this is written as

$$\max_{x \in S} c^T x = -\min_{x \in S} (-c^T x)$$

Example: $\max(c^T x) = 50$, thus $\min(-c^T x) = -50$. Therefore $50 = \max(c^T x) = -(\min(-c^T x)) = -(-50) = 50$

If we are referring to the optimal argument/solution vector x^* (argmax v.s. argmin), rather than the optimal value

$$\arg \max_{x \in S} c^T x = \arg \min_{x \in S} (-c^T x)$$



Convert to Standard form

$$\begin{aligned} & \min 2x_1 + 4x_2 \\ & \text{s.t. } x_1 + x_2 \geq 3 \\ & \quad 3x_1 + 2x_2 = 14 \\ & \quad x_1 \geq 0 \end{aligned}$$

$$\text{Let } x_2 = x_2^+ - x_2^-, \quad x_2^+, x_2^- \geq 0$$



$$\begin{aligned} & \min 2x_1 + 4x_2^+ - 4x_2^- \\ & \text{s.t. } x_1 + x_2^+ - x_2^- - s_1 = 3 \\ & \quad 3x_1 + 2x_2^+ - 2x_2^- = 14 \\ & \quad x_1, x_2^+, x_2^-, s_1 \geq 0 \end{aligned}$$

※ Resource - Allocation Problem: Wyndor Glass Co.

- Plant 1 → produces Aluminum frames
 - Plant 2 → produces wood frames
 - Plant 3 → produces glass and assembles the products
- ◆ Product 1: An 8-foot glass door aluminum framing ⇒ Plants 1 & 3
 ◆ Product 2: A 4 × 6 foot double-hung wood-framed window ⇒ Plants 2 & 3

Data for Wyndor Glass Co. Problem

Data:

	Production time per batch hours		Production Time available per Week, hours
	PRODUCT		
Plant	1	2	
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	\$3,000	\$5,000	

Let x_1 = number of batches of product 1 produced per week
 x_2 = number of batches of product 2 produced per week

$$\begin{aligned}
 & \text{(LP)} \quad \max \quad 3x_1 + 5x_2 \quad (\text{Profit}) \\
 & \text{s.t.} \quad x_1 \leq 4 \\
 & \quad \quad 2x_2 \leq 12 \\
 & \quad \quad 3x_1 + 2x_2 \leq 18 \\
 & \quad \quad x_1 \geq 0, x_2 \geq 0
 \end{aligned}$$

※ Graphical Representation and Solution

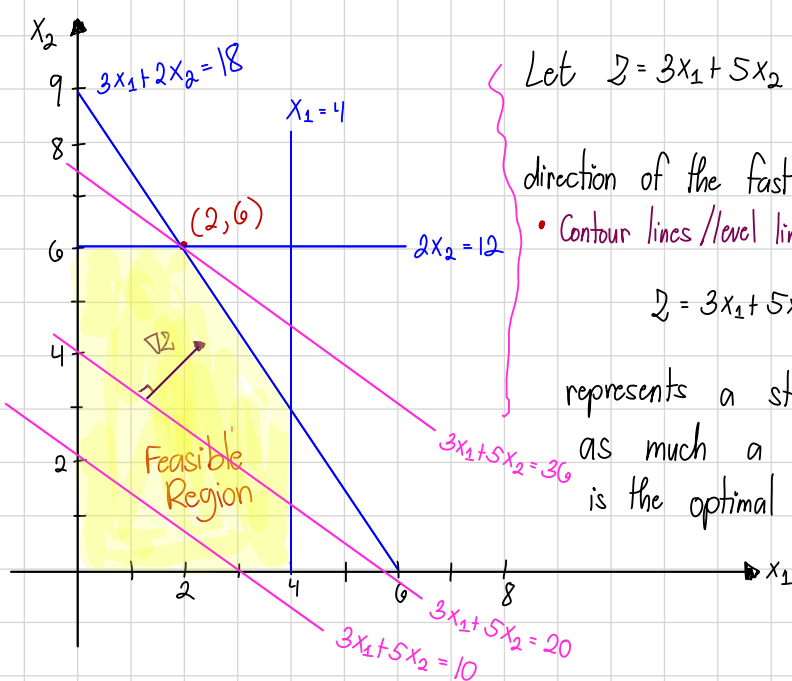
Theorem: Suppose f is a differentiable function of two or three variables. The maximum value of the directional derivative $D_u f(x)$ is $|\nabla f(x)|$ and it occurs when the unit vector u has the same direction as $\nabla f(x)$.

Proof:

$$D_u f = \nabla f \cdot u = |\nabla f| |u| \cos \theta = |\nabla f| \cos \theta$$

where θ is the angle between ∇f & u . Since $\max_{\theta \in \mathbb{R}} \cos \theta = 1$ and this occurs when $\theta = 0$. Therefore the maximum value of the $D_u f$ is $|\nabla f|$.

Remark: ∇f gives the direction of the fastest increase of f , $-\nabla f$ gives the direction of the fastest decrease of f .



Let $z = 3x_1 + 5x_2$ then $\nabla z = \begin{bmatrix} \frac{\partial z}{\partial x_1} \\ \frac{\partial z}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and this gives us the direction of the fastest increase.

• **Contour lines / level lines:** For any fixed $z \in \mathbb{R}$, the equation

$$z = 3x_1 + 5x_2 \Rightarrow x_2 = -\frac{3}{5}x_1 + \frac{z}{5}$$

represents a straight line with slope $m = -\frac{3}{5} \perp \nabla z$. Move the line as much as possible in the direction of ∇z . The vector $x = (2, 6)$ is the optimal solution

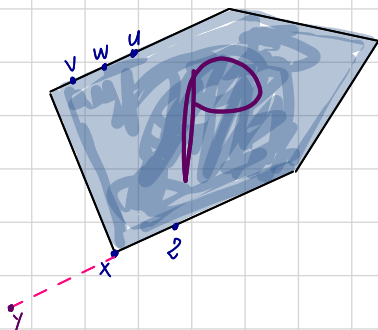
Conclusion: $x_1 = 2$, $x_2 = 6 \Rightarrow z = 36$. Produce products 1 and 2 at a rate of 2 batches per week and 6 batches per week, respectively, with a resulting profit of \$36,000 per week.

Here, each **constraint boundary** is a line that forms the boundary of what is permitted by the corresponding constraint. The points of intersection are the **corner-point solutions** of the problem.

* Some Geometry of Linear Programming

Definition: A **polyhedron** is a set that can be described in the form $\{x \in \mathbb{R}^n \mid Ax \leq b\}$, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

Definition: Let P be a polyhedron. A vector $x \in P$ is an **extreme point** of P if we **CANNOT** find two vectors $y, z \in P$, $y \neq x$, $z \neq x$ and a scalar $\lambda \in [0, 1]$ such that $x = \lambda y + (1-\lambda)z$.

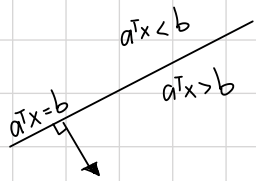


- w is NOT an extreme point because it is a convex combination of v and u .
- x is an extreme point: if $x = \lambda y + (1-\lambda)z$ and $\lambda \in [0, 1]$, then either $y \notin P$ or $z \notin P$, or $x = y$, or $x = z$.

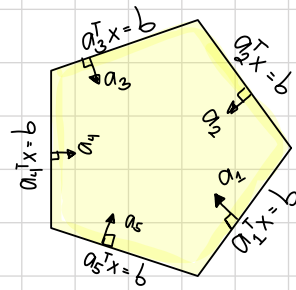
Definition: Let a be a nonzero vector in $\mathbb{R}^n$ and let b be a scalar.

- The set $\{x \in \mathbb{R}^n \mid a^T x = b\}$ is called a **hyperplane**.
- The set $\{x \in \mathbb{R}^n \mid a^T x \leq b\}$ is called a **half space**.

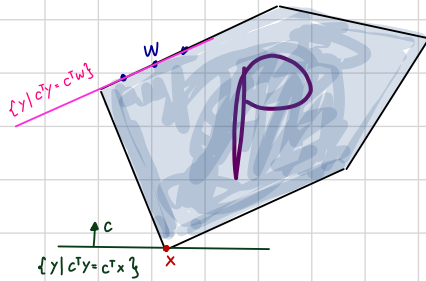
* Some Geometry of Linear Programming



A hyperplane and two half spaces

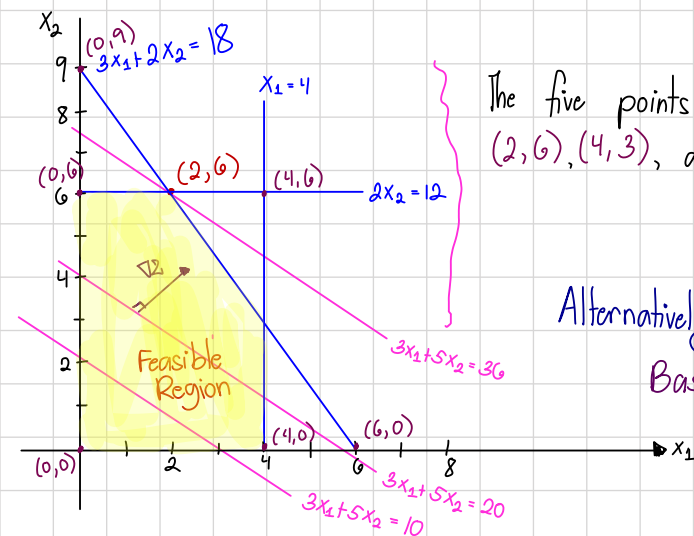


The polyhedron $\{x \mid a_i^T x \geq b_i, i=1, \dots, 5\}$ is the intersection of five halfspaces.



The line at the bottom touches P at a single point and x is a vertex. While w is not a vertex because there's no hyperplane that meets P only at w .

* Going back to the Wyndor Problem (Graphical Solution)



The five points that lie on the corners of the feasible region: $(0,0)$, $(0,6)$, $(2,6)$, $(4,3)$, and $(4,0)$ — are corner point feasible solutions (CPF).

CPF $\equiv$ Vertex $\equiv$ Extreme Point

Alternatively,

Basic Feasible Solution (BFS) $\iff$ Corner Point Feasible Solution

Basic Solution $\iff$ Corner Point Solution

Fundamental Theorem of Linear Programming: If a linear programming problem has a solution, it must occur at a vertex of the set of feasible solutions.

Theorem: Consider the constraints $Ax=b$ and $x \geq 0$, assume that $A \in \mathbb{R}^{m \times n}$ has linearly independent rows. A vector $x \in \mathbb{R}^n$ is a basic solution if and only if $Ax=b$, and there exists indices $B(1), \dots, B(m)$ such that:

- (a) The columns $A_{B(1)}, \dots, A_{B(m)}$ are linearly independent
- (b) If $i \neq B(1), \dots, B(m)$, then $x_i = 0$.

Definition: Let V be a vector space over a field $\mathbb{F}$ and $v_1, \dots, v_n \in V$. If for all $a_1, \dots, a_n \in \mathbb{F}$

$$\sum_{i=1}^n a_i v_i = \vec{0} \quad \text{implies} \quad a_1 = \dots = a_n = 0$$

we say $\{v_1, v_2, \dots, v_n\}$ is linearly independent.

If x is a basic solution then

- $x_{B(1)}, \dots, x_{B(m)}$ are called basic variables; the remaining variables are nonbasic.
- $A_{B(1)}, \dots, A_{B(m)}$ are called basic columns

$B = \begin{bmatrix} | & | & \dots & | \\ A_{B(1)} & A_{B(2)} & \dots & A_{B(m)} \\ | & | & \dots & | \end{bmatrix}$; $x_B = \begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix}$; here $B \in \mathbb{R}^{m \times m}$ and is invertible which means that $B^{-1}B = I_m$ where I_m is an identity matrix. B is invertible or nonsingular since it is a square matrix ($m \times m$) and has a full rank ($\text{rank}(B) = m$).

Basis Matrix

The basic variables

* Solving Linear Programming: The Simplex Method

Simplex Method a general procedure for solving linear programming problems. Developed by George Dantzig in 1947.

Consider again the following LPP

$$\begin{aligned} \max \quad & 3x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 \leq 4 \\ & 2x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

To Standard form $\Rightarrow$

$$\begin{aligned} \min \quad & -3x_1 - 5x_2 \\ \text{s.t.} \quad & x_1 + s_1 = 4 \\ & 2x_2 + s_2 = 12 \\ & 3x_1 + 2x_2 + s_3 = 18 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0 \end{aligned}$$

$$\min -3x_1 - 5x_2 + 0s_1 + 0s_2 + 0s_3$$

$$C = \begin{bmatrix} -3 \\ -5 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow C^T = \begin{bmatrix} x_1 & x_2 & s_1 & s_2 & s_3 \\ -3 & -5 & 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 & x_2 & s_1 & s_2 & s_3 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

An iteration of the Simplex Method

1.) Start with a basis matrix, consisting of columns $A_{B(1)}, \dots, A_{B(m)}$, and an associated basic feasible solution x

Iteration 0:

$$x_B = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \text{ \& } x_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, C_B^T = [0 \ 0 \ 0] ; B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow x_B = B^{-1}b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}, x_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

3x3 3x1

Current CPFS: $x = [0 \ 0 \ 4 \ 12 \ 18]^T$

2.) Compute for the reduced cost $\bar{c}_j = c_j - c_B^T B^{-1} A_j$ for all nonbasic indices j . If all $\bar{c}_j \geq 0$, the current basic feasible solution is optimal, and the algorithm terminates; else choose some j for which $\bar{c}_j < 0$.

$$\bar{c}_1 = c_1 - c_B^T B^{-1} A_1 = -3 - [0 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = -3 - 0 = -3$$

$$\bar{c}_2 = c_2 - c_B^T B^{-1} A_2 = -5 - [0 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = -5 - 0 = -5 ; \text{ Choose as our entering variable}$$

3.) Compute $u = B^{-1}A_j$. If no component u is positive, we have $\theta^* = \infty$, the optimal cost is $-\infty$, and the algorithm terminates.

$$u = B^{-1}A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \begin{matrix} \rightarrow u_{B(1)} \\ \rightarrow u_{B(2)} \\ \rightarrow u_{B(3)} \end{matrix}$$

4.) If some component u is positive, let

$$\lambda^* = \min_{\{i=1, \dots, m \mid u_i > 0\}} \frac{x_{B(i)}}{u_i}$$

$$\lambda^* = \min \left\{ \frac{x_{B(2)}}{u_2}, \frac{x_{B(3)}}{u_3} \right\} = \min \left\{ \frac{12}{2}, \frac{18}{2} \right\} = \min \left\{ \frac{12}{2}, \frac{18}{2} \right\} = \min \{6, 9\} = 6$$

5.) Let l be such that $\lambda^* = \frac{x_{B(l)}}{u_l}$. Form a new basis by replacing $A_{B(l)}$ with A_j . If y is the new basic feasible solution, the values of the new basic variables are $x_j = \lambda^*$ and $x_{B(i)} = \lambda^* u_i$, $i \neq l$

• The minimum is achieved at $i=2$, $A_{B(2)}$ i.e., s_2 . Thus $l=2$, and s_2 leaves the basis

$$x_N = \begin{bmatrix} x_1 \\ s_2 \end{bmatrix} \quad \& \quad x_B = \begin{bmatrix} s_1 - \lambda^* u_{B(1)} \\ x_2 = \lambda^* \\ s_3 - \lambda^* u_{B(3)} \end{bmatrix} = \begin{bmatrix} 4 - 6(0) \\ 6 \\ 18 - 6(2) \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 6 \end{bmatrix}; \text{ New CPFS: } x = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ 4 \\ 6 \end{bmatrix}$$

Iteration 1:

$$\bullet B = [A_3 \ A_2 \ A_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow B^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad c_B^T = [0 \ -5 \ 0]$$

$$\bullet \bar{c}_1 = c_1 - c_B^T B^{-1} A_1 = -3 - [0 \ -5 \ 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = -3 - 0 = -3$$

$$\bullet \bar{c}_4 = \bar{c}_4 - c_B^T B^{-1} A_4 = 0 - [0 \ -5 \ 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0 - \left(-\frac{5}{2}\right) = \frac{5}{2}$$

Since $\bar{c}_3 < 0$, the solution is not optimal. Choose $j=1$, i.e. x_1 as the entering variable

$$u = B^{-1}A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

$$\lambda^* = \min \left\{ \frac{x_{B(1)}}{u_{B(1)}}, \frac{x_{B(3)}}{u_{B(3)}} \right\} = \min \left\{ \frac{4}{1}, \frac{6}{3} \right\} = \min \{4, 2\} = 2; \quad s_3 \text{ is the leaving variable}$$

$$x_N = \begin{bmatrix} s_3 \\ s_2 \end{bmatrix}; \quad x_B = \begin{bmatrix} s_1 - \lambda^* u_{B(1)} \\ x_2 - \lambda^* u_{B(2)} \\ x_1 = \lambda^* \end{bmatrix} = \begin{bmatrix} 4 - 2(1) \\ 6 - 2(0) \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix}; \text{ New CPFS: } x = \begin{bmatrix} 2 \\ 6 \\ 2 \\ 0 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ s_1 \\ s_3 \end{matrix}$$

Iteration 3:

$$\bullet B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow B^{-1} = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}; \quad c_B^T = [0 \ -5 \ -3]$$

$$\bullet \bar{c}_4 = c_4 - c_B^T B^{-1} A_4 = 0 - [0 \ -5 \ -3] \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0 - \left(-\frac{3}{2}\right) = \frac{3}{2} > 0$$

$$\bullet \bar{c}_5 = c_5 - c_B^T B^{-1} A_5 = 0 - [0 \ -5 \ -3] \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0 - (-1) = 1 > 0$$

Since all $\bar{c}_j > 0$, the current CPFS is optimal. Therefore, $x_1^* = 2$ & $x_2^* = 6$ and max profit = $3x_1^* + 5x_2^* = 3(2) + 5(6) = 6 + 30 = 36$.